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Comparison of a novel rapid sampling method to serum and tonsil scraping to detect PRRSV in acutely infected sows

Peng Li [a], Ana Paula Serafini Poeta Silva [a], Daniel Carnevale de Almeida Moraes [a], Paul Yeske [b], Onyekachukwu Henry Osemeke [a], Edison Souza Magalhães [a], Gustavo De Sousa E Silva [a], Daniel Correia Lima Linhares [a,*]

[a] Department of Veterinary Diagnostic and Production Animal Medicine, Ames, IA 50010, United States
[b] Swine Vet Center, Saint Peter, MN 56082, United States

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ABSTRACT

Few practical methods are available to monitor the PRRSV status of the sows. Common sampling methods for sows like serum sampling, and tonsil scraping involve restraining individual sows and are labor-intensive, time-consuming, relatively invasive, and therefore, have limited use in large-scale production settings. Thus, a practical and rapid method of sampling large numbers of sows is needed. This study aimed to develop a new sampling method, named tonsil-oral scraping (TOSc) and compare TOSc to serum and tonsil scraping in terms of PRRSV qPCR detection rate and Ct values in thirty matched sows, thirty days after PRRSV outbreak. TOSc recovered a mixture of oral fluids and tonsil exudates from the sow oral cavity within seconds without restraining the animals. Results showed that, numerically, the TOSc samples had higher PRRSV qPCR detection rate (100 %) compared to serum (16.8 %) and tonsil scraping (73.1 %). Moreover, TOSc samples had lower average Ct values (29.7) than tonsil scraping (30.7) and serum (35.2). There was no significant difference in the detection rate between TOSc and tonsil scraping (Tukey test, $p = 0.992$), while there was a significant difference between serum and tonsil scraping (Tukey test, $p < 0.001$), as well as between serum and TOSc (Tukey test, $p < 0.001$). In terms of Ct values, there was no statistically significant difference between TOSc and tonsil scrapings (Dunn Test, $p > 0.05$), while there was a significant difference between tonsil scraping with serum (Dunn Test, $p < 0.01$), and TOSc with serum (Dunn Test, $p < 0.01$). Our results suggest great potential of the TOSc as a novel, practical, and rapid tool for PRRSV RNA detection in sows to assess sow herd status.

1. Introduction

There are several well-established population-based sample types, such as processing fluids (PF), family oral fluids (FOF), and tongue tip fluids (TTF) to monitor porcine reproductive and respiratory syndrome virus (PRRSV) activity in suckling pigs (Almeida et al., 2020; Baliellas et al., 2021; Lopez et al., 2022; Lopez et al., 2020; Machado et al., 2022; Vilalta et al., 2018). The sows are a major source of PRRSV infection to piglets (Pileri and Mateu, 2016). However, these aggregated samples originated from the suckling piglet sub-population. While the population-based sample types enable sampling more animals providing increased herd sensitivity, they are not useful to provide individual information for estimating prevalence. Moreover, samples originating from suckling piglets cannot directly determine individual sow status for PRRSV. Therefore, an effective and practical tool to directly sample sows is needed.

Commonly used sampling methods for the sows to detect pathogen include serum and tonsil scraping (Nielsen et al., 2005; Tousignant et al., 2017; Wills et al., 2003). However, serum and tonsil scraping are time consuming and labor-intensive for large screening especially in low prevalence scenarios where relatively large sample sizes are required to detect at least one positive animal. Moreover, both methods require the restraining of sows, which causes stress and requires skilled labor.

In this study, we developed a novel tonsil oral scraping (TOSc) method which collects fluids from the oral cavity and tonsil. The method was developed by adapting a similar sow sampling method used for the test-and-removal of sows infected with African swine fever virus (ASFV) in China (Li, 2022). Thus, the objective of this study was to compare the

* Corresponding author.
E-mail address: linhares@iastate.edu (D.C.L. Linhares).

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TOSc sample with serum and tonsil scraping in terms of PRRSV RT-qPCR detection rate and average Ct values in acutely infected sows.

2. Methods

2.1. Study design and target population

A cross-sectional study was performed in a 2400 breed-to-wean herd seeking to eliminate PRRSV following a recent outbreak, and located in Minnesota, US. Thirty sows in the farrowing crates were conveniently selected and sampled for PRRSV qPCR testing. Serum, tonsil scraping, and TOSc were collected from each of the 30 sows. The institutional Animal Care and Use Committee IACUC of [de-identified as per PVM request].

2.2. Sample collection

2.2.1. TOSc collection

The TOSc tool was constructed using an artificial insemination (AI) rod (GoldenPig Catheter, Model 320100 from IMV international, US), a rubber thimble (Large Finger Pad, Model 098130 from Staples Inc, US) covered by a layer of cotton gauze (Model 074144, from Honeywell Inc, US), to make it more abrasive, and a rubber band to attach the cotton gauze to the AI rod (Fig. 1).

TOSc was collected without restraining the sows (Fig. 2). The head of the AI rod was placed beside the mouth of the sow to attract her attention. The rod was then inserted into the sow's mouth, with the help of a headlamp, directed toward the tonsil area with an upwards angle, and scrubbed back and forth for ten seconds. The qualified sample was defined as viscous and mucous-like. The head part of the rod with sample on it was cut by a sterilized clip and then transferred to a 50 ml conical tube (Corning Science Mexico S.A. de C.V., Tamaulipas, Mexico) with two ml of PBS. An average of 0.91 ml (SD=0.31 ml) TOSc samples were collected.

2.2.2. Tonsil scraping

After collecting the TOSc samples, the sows were restrained using a snare, and conventional tonsil scraping samples were collected for each sow, as described previously (Nielsen et al., 2005; Tousignant et al., 2017; Wills et al., 2003). Briefly, the sows were restrained with a snare, and the mouth was held open with a speculum (metal mouth gag). A long-handled metal spoon was used to scrape the palatine tonsil (as done for laryngeal swabs). The spoon was carefully inserted with the bowl side up, avoiding touching the tongue or other areas. The sample on the spoon was transferred using a disposable Puritan sterile polyester tipped applicator (Puritan Medical Products Company LLC, Guilford, ME, USA) to a five ml Falcon tube (RPI research products international, Mt. Prospect, IL, USA) containing one ml of PBS. The spoon was decontaminated with disinfecting wipes (Clorox® Disinfecting Wipes, Clorox Company, US) and then dried with disposable paper towel.

2.2.3. Serum collection

Precaval vein blood was also collected using B.D. Vacutainer® (Becton Dickinson and Company, Franklin Lakes, NJ, USA) after the tonsil scrapings with the sows snared and restrained. After collection of the whole blood, the blood was allowed to clot by leaving it undisturbed at room temperature for 15–30 min. Serum was then separated from the clot by centrifugation at 1500xg for 5 min.

2.3. Diagnostic testing

A total of 30 serums, 30 tonsil scraping samples and 30 TOSc samples were chilled and shipped on ice to the Iowa State University Veterinary Diagnostic Laboratory (ISU-VDL). All samples were individually tested for PRRSV RNA detection by RT-qPCR using validated protocols.

2.4. Statistical analysis

Descriptive statistics reported include the PRRSV RT-qPCR detection rate and distribution of Ct values for each sample type. Bayesian GLM model was used to model the logit of detection rate as a function of different sample types, and differences in detection rates among different sample types were assessed. Specifically, Tukey-Kramer test was used as a post hoc test to compare the marginal means of detection rates. The Dunn's Test (non-parametric test) was performed to assess if there was a difference in the Ct values between the sample types. All analyses were performed using the R program 4.2.2 (R Core Team, 2019).

3. Results

3.1. PRRSV RNA detection rate

As shown in Table 1 and Fig. 3, TOSc had the higher detection rate of PRRSV RNA detection rate of 99.6 %, with a 95 % Credibility interval (CI) between 95.1 %–100 %, compared with serum 16.7 % (95 % CI 5.7 %–30.7 %), and tonsil scraping 73.5 % (95 % CI 57.3 %–87.6 %). However, as shown in Table 1, there was no significant difference in the detection rate between TOSc and tonsil scraping (Tukey test, $p = 0.993$), while there was a significant difference between serum and tonsil scraping (Tukey test, $p < 0.001$), as well as between serum and TSc (Tukey test, $p < 0.001$).

3.2. Ct values

TOSc showed numerically lower average Ct values (29.7) than tonsil scraping (30.7) and serum (35.2). There was no statistically significant difference between TOSc and tonsil scrapings (Dunn Test, $p > 0.05$), while there was a significant difference between tonsil scraping with serum (Dunn Test, $p < 0.01$), and TOSc with serum (Dunn Test, $p < 0.01$).

3.3. Discussion

In this study a tool was developed and named a TOSc collector (Fig. 1), which may be used to collect a composite of oral fluids and tonsil exudate. It was successfully used to collect samples from each sow within 30 s without the necessity of restraint. The TOSc was modeled after a sample collection tool reported by Newhope Swine Research Institute for the test-and-removal of individual ASFV-infected sows in China (LI, 2022). This tool is user-friendly and animal friendly as it does not require restraining the sows (Fig. 2) and thuscan allow for larger numbers of animals to be tested as well.

The specific modification made to the TOSc collector for PRRSV

Fig. 1. TOSc collector prototype adapted by our group. Zoom of head part of the collector(a) and overall picture of the TOSc collector (b).

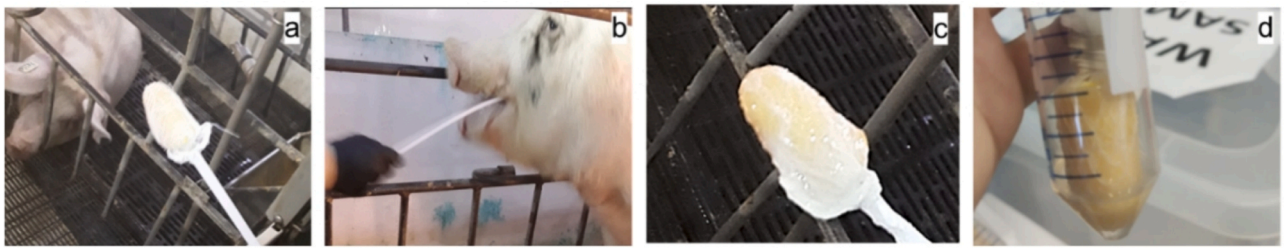

Fig. 2. TOSc sampling in the acutely infected farm in Minnesota. TOSc collector before collection (a), sample collection (b), TOSc samples after collection(c), TOSc fluid after processing(d).

Table 1
PRRSV detection rate and average Ct values of three different sample types in 30 sows.

Sample type	Serum	Tonsil Scraping	Tonsil Oral scraping (TOSc)
PRRSV detection rate [95 % Credibility Interval]	^{a}16.7% [5.7 %,30.7 %]	^{b}73.1% [57.3 %, 87.6 %]	99.6 %^{b} [95.1 %, 100 %]
Average Ct values and range in positive samples (lowest value to highest value)	35.2 [33.6–36.4]	30.7 [25.1–35.4]	29.7 [24.1–36.3]

ab:Different superscript letters indicate significant differences in least square means (Tukey test, $p < 0.05$)

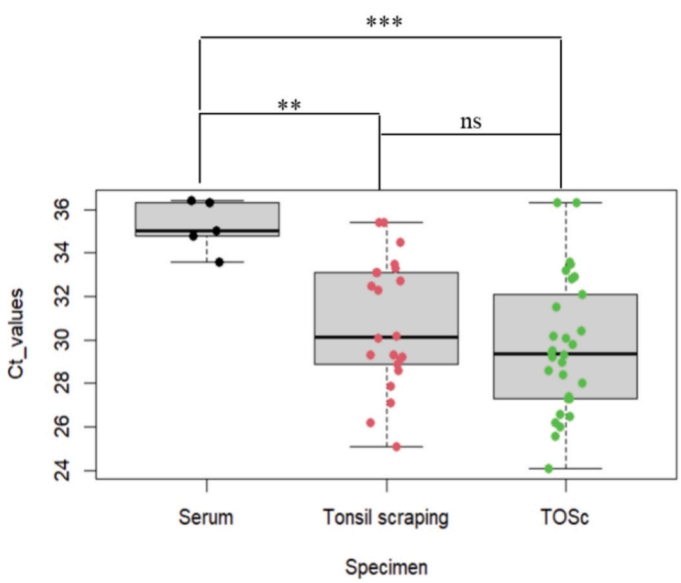

Fig. 3. Distribution of PRRSV Ct values of positive samples from three different sample types.
Ct≤37 was deemed positive; TOSc: Tonsil Oral Scraping; ***, $p < 0.001$; **, $p < 0.01$; ns: not significant, $p > 0.05$ (Dunn Test).

detection was to add a rubber thimble between the head of an AI rod and an outer layer of cotton gauze to make the tool more abrasive to mimic the conventional tonsil scraping process, as tonsil scrapping provides superior sensitivity compared to serum and other sample types to detect long-term PRRSV carrier pigs due to virus replication in tonsil at this phase of infection (Horter et al., 2002).

There was no significant difference in terms of detection rate and Ct values between tonsil scraping and TOSc. However, in our study, the TOSc had numerically higher PRRSV RNA detection rate and average lower Ct values even than tonsil scraping samples in the acute phase of PRRSV infection although statistically there is no significant difference. We hypothesized that this phenomenon was possibly because TOSc recovers a mixture of oral fluid and tonsil exudate. Thirty days after PRRSV exposure, both oral fluids and tonsil tissue contain high viral loads (Henao-Diaz, Ji et al., 2020; Lopez and Osorio, 2004). Another possible explanation might be, normally TOSc yields one to two ml of fluids, while conventional tonsil scraping yields only approximately 100 μl of fluids. Moreover, conventional tonsil scraping method which requires restraint of the sow, yields a more inconsistent sample. In contrast, TOSc yields a more consistent sample from sows.

One major problem with PRRSV control and elimination in the pig industry today is the persistence of PRRSV with the circulating strains in the herd. It is not uncommon for producers to report a rebreak of the same PRRSV strain while in the process of PRRSV elimination (Linhares et al., 2014). This indicates that PRRSV is being actively in the herd somewhere. This could be partly due to the tropism of PRRSV for pig immune cells (Lunney et al., 2016) and thus residing in the host for a relatively long period of time before being cleared. The other reason that might explain the persistence of PRRSV is a large herd size, which means larger susceptible populations for the virus to replicate. This creates difficulties for sample collection strategies to identify PRRSV status. Higher level of viral shedding of current strains, such as PRRSV 1–4–4 L1C variants, was also believed to contribute to the persistence (Kikuti et al., 2021; Trevisan et al., 2021). Commonly used sample types for the sows to detect various pathogens include serum and tonsil scraping (Nielsen et al., 2005; Tousignant et al., 2017; Wills et al., 2003) and occasionally, oral fluid (OF) (Brent Pepin, 2015; Pol et al., 2017). However, serum and tonsil scraping are time consuming and labor-intensive for screening a large number of animals, especially in the low prevalence scenario. Moreover, both methods require the restraint of the sows, which causes stress impacting animal welfare. Oral fluid is most frequently used for population-based sampling purposes (Henao--Diaz, Gimenez-Lirola et al., 2020; Pol et al., 2017). When used to collect samples from individual sows. the success rate, varied from 14.6 % to 67.4 % (Brent Pepin et al., 2015). Unlike oral fluid, often used as an aggregated sample type, TOSc recovers individual samples of a mixture of oral fluid and tonsil exudate. The speed and ease of use of TOSc will potentially open doors of research to better understand of dynamics of individual sow PRRSV status as well.

Altogether, TOSc developed in this study has potential for large scale herd screening purposes for PRRSV detection, surveillance, control, and elimination in the field, especially in herds that are having difficulty reaching their goals.

Some limitations of this study include the relatively small sample size and the early phase of PRRSV infection in the sampled sows, and more work needs to be done in herds that are in late closure to assess the viability of this method More variability in PRRSV RNA detection in the various sample types obtained from the sow is expected over time as immunity builds (Henao-Diaz, Ji et al., 2020; Lopez and Osorio, 2004). At different PRRSV infection stages, different samples have different detection rate (Henao-Diaz, Ji et al., 2020). In a review by Henao-Diaz, at early phases of PRRSV infection, it is easy to detect virus from blood and oral fluids or nasal secretions. The viremia in adult animals last approximately one month, while the virus can still be detected in secretions for three months. However, as the disease progresses, even

though the virus is not detected from oral fluids or nasal secretions, the virus can still be detected in tonsil scrapings (Pileri and Mateu, 2016). Since TOSc recovers a mixture of tonsil exudate and oral fluids, may be interesting to explore the performance of TOSc at different stages of PRRSV infection and using larger sample sizes by comparison TOSc with serum, oral fluid, and tonsil scrapings.

4. Conclusion

The TOSc samples had a superior PRRSV RNA detection rate and lower PCR Ct values compared to serum and tonsil samples.

CRediT authorship contribution statement

Edison Souza Magalhães: Methodology, Resources. **Osemeke Onyekachukwu:** Formal analysis, Resources, Visualization, Writing – review & editing. **Yeske Paul:** Investigation, Project administration, Resources, Supervision, Writing – review & editing. **Linhares Daniel Correia Lima:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Writing – review & editing. **Silva Gustavo:** Methodology, Project administration, Resources, Supervision, Writing – review & editing. **de Almeida Moraes Daniel:** Investigation, Methodology, Resources, Writing – review & editing. **Silva Ana Paula Serafini Poeta:** Formal analysis, Investigation, Methodology, Resources, Writing – review & editing. **Li Peng:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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